Original Article

Leveraging drone technology in health systems strengthening: A cost-effectiveness analysis from Northeast India

Prakamya Gupta[1], Ragini Bhatia[1], Ranjan Kumar Choudhury[1], Atul Kotwal[2]

[1]*Division of Healthcare Technology, National Health Systems Resource Center, New Delhi, India*
[2]*Executive Director, National Health Systems Resource Center, New Delhi, India*

Abstract

Introduction: Unmanned aerial vehicles (UAVs) or drones have the potential to revolutionize healthcare logistics last-mile delivery. The use of drone technology in healthcare is relatively a new domain with a few sites undertaking pilot trials resulting in a lack of data for analysis, particularly in the healthcare sector. The present study aims to explore the longitudinal analysis and cost-effectiveness assessment of the implementation of drone technology in the public healthcare system. **Methods:** Retrospective data from December 2022 to January 2024 were obtained from the World Bank sponsored Meghalaya Drone Delivery Network under the Meghalaya Health Systems Strengthening Project. An economic evaluation of the drones versus vehicle transportation of medical logistics from hub to spoke was conducted. Carbon emissions from vehicles and drones for transportation were also compared. **Results:** A total of 214 flights were conducted, transporting 199·05 kg of payload to 11 spoke sites (primary healthcare centers) during 7 months period. The average cost by road transportation for one round trip was INR 2902 (US$ 34·80), whereas by drone, it was INR 3743 (US$ 44·87). The Incremental Cost-Effectiveness Ratio (ICER) for drone transportation is INR 12 (US$ 0·14) per Disability Adjusted Life Years (DALY) averted as compared to vehicle transportation of medical logistics. The healthcare facility with drone services was able to avert 106 DALYs compared to 36 DALYs per month by the healthcare facility with a conventional mode of transportation. The drone services would require an additional INR 56 (US$ 0·67) to save 1 minute to transport medical logistics. The CO_2 emission per payload delivered by drones was up to 99% less than the conventional mode of transportation. **Conclusion:** Drone technology in healthcare complements the existing supply chain routes and is a value addition for saving lives in unforeseen medical emergencies. Further trials are required for firming up use cases to understand the challenges associated.

Keywords: Carbon emission, cost-effectiveness, drones, healthcare, India, Meghalaya

Introduction

Unmanned aerial vehicles (UAVs) or drones have proliferated in various domains because of their ability to traverse through difficult terrains, reducing time and labor costs. They have been used for surveillance and humanitarian aid delivery and are now being increasingly used to transport medical supplies and diagnostic samples. Timely delivery of medical supplies is essential in the healthcare sector for improved health outcome. The conventional supply chain is often hampered by numerous factors such as poor transportation networks, traffic, and adverse environmental conditions. Drone operations can leapfrog the last mile logistics solution for transporting medical supplies in hard-to-reach difficult hilly terrains.[1] The drones have been explored for their rapid and safe delivery of medical supplies.[2]

Address for correspondence: Dr. Prakamya Gupta, National Health Systems Resource Centre, Munirka, New Delhi - 110 067, India.
E-mail: prakamya1986@gmail.com

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Drones have significant potential to supplement the existing medical logistics delivery supply chain. Drones can prove to be last-mile logistics solutions for transporting emergency medical supplies in hard-to-reach difficult hilly terrains. However, drone-based healthcare logistics is more challenging compared to agricultural drones in terms of distance covered, payload, regulatory approvals, and so on.[3] Despite several flight trials and pilot studies of using drone technology in delivering medical supplies, limited evidence is available regarding the cost-effectiveness[4-6] in the integration of drone technology with the existing supply chain routes and its impact in the long term.

Meghalaya, a hilly state in North-East India, is characterized by highland plateaus and valleys interspersed with various geological features. The rugged terrain and poor road infrastructure have made it difficult to deliver medical supplies in the north-eastern state of Meghalaya, India. The Meghalaya Government, through the Meghalaya Health Systems Strengthening Project (MHSSP), in December 2022 initiated drone-based medical supplies delivery to mitigate the frequent stockout of essential and life-saving drugs in a limited scale. The MHSSP drone delivery supplied emergency medicines to far-flung and remotely located Primary Healthcare Centres (PHCs) with carriage of diagnostic samples as and when required in reverse logistics model. The present study explores the cost-effectiveness with health outcomes of the use of drones in the public health system in Meghalaya.

Methodology

Under the MHSSP, a hub and spoke model was considered to cover maximum healthcare centers in a radius of 60 km. The Sub Divisional Hospital (SDH) - Jengjal (SDH Jengjal) (Hub) located at 536 m/1760 ft above mean sea level is a 43-bedded hospital with a gynecologist and medical specialist, a functional ICU, a labor room, HDU, and Neonatal ICU. PHCs located within the operational radius were selected for medical supplies delivery using drones, based on the remoteness/difficult approach roads in the West Garo Hill district of Meghalaya. The total number of flights and payloads delivered from the hub to the spokes (PHC) from December 2022 to January 2024 were analyzed.

Drone specifications

The MHSSP engaged a drone service provider (M/s TechEagle Pvt Ltd) for the supply of emergency medicines to far-flung and remotely located PHC with carriage of diagnostic samples as and when required in reverse logistics. The service provider provided services in the Tura district Meghalaya with the SDH Jengjal as Hub. The Vertiplane X3 drone was a Beyond Visual Line of Sight (BVLOS) category vertical take-off and landing (e-VTOL) drone that had a maximum range of 100 km, an average cruise speed of 90 km/h, an operating altitude of 400 ft above ground level, and a wind resistance of 45 km/h. The drone used a lithium polymer battery with a capacity of 22,000 mAh and 3·6 V and a payload carrying capacity 2–5 kg. All flights were operated by remote pilot training organization-certified drone pilots. Necessary drone flight permissions were obtained by the drone service provider.

Climate data collection

The state of Meghalaya receives high rainfall with around 236·74 wet days (64·86% of the total) and 301·2 mm (11·86 inches) of precipitation on average each year. Average monthly precipitation, number of rainy days per month, and average wind speed in Jengjal region, Meghalaya, from December 2022 to January 2024 were acquired from the World Weather Online.[7] The good and bad weather days were compared with the flight data.

Cost-effectiveness analysis

An economic evaluation (activity-based costing) of drone versus vehicle-based transportation of medical logistics from hub to spoke was conducted from the provider's perspective (public health system). A monthly cycle for economic evaluation with a lifetime horizon and 3% discount rate for costs was used in the cost-effectiveness model. No discount rate was applied for outcomes or consequences. A bottom-up approach was used for cost estimation. Two sites were selected, one interventional site with drone services (PHC Pedaldoba) and one comparator spoke without drone services (PHC Bajengdoba). A retrospective cross-sectional study design was employed. Data were collected and recorded from the health facility records for the past 1 year and from interviews of the staff and healthcare workers. Vehicle and drone operational data were obtained from the administration at the healthcare facility of the Hub and Spokes. The data on the frequency and type of consumable transportation between Hub and Spoke in a year were recorded. The average cost of a round trip, the flow chart of the procedure, the HR cost, the fuel consumption, and the maintenance cost were then calculated. The decision tree model for the cost-effectiveness analysis, illustrated in Figure 1, systematically maps the clinical and operational pathways of both the intervention facility (with drone services) and the comparator facility (without drone services), enabling a comprehensive estimation of costs and outcomes to effectively compare their economic and health-related impacts.

Incremental Cost Effectiveness Ratio (ICER) was calculated, which is the difference between cost-effectiveness of intervention and comparator. For calculation of cost incurred each year, the cost was multiplied by the annualization factor (AF).

Formula for AF:

$$AF = \frac{r(1+r)^{L}}{(1+r)^{L}-1}$$

r = discount rate

L = Life of equipment

AF = 0·117 when the discount rate is 3% and the life of the equipment is 10 years.

The cost is then multiplied by AF to get Equivalent Uniform Annual Cost.

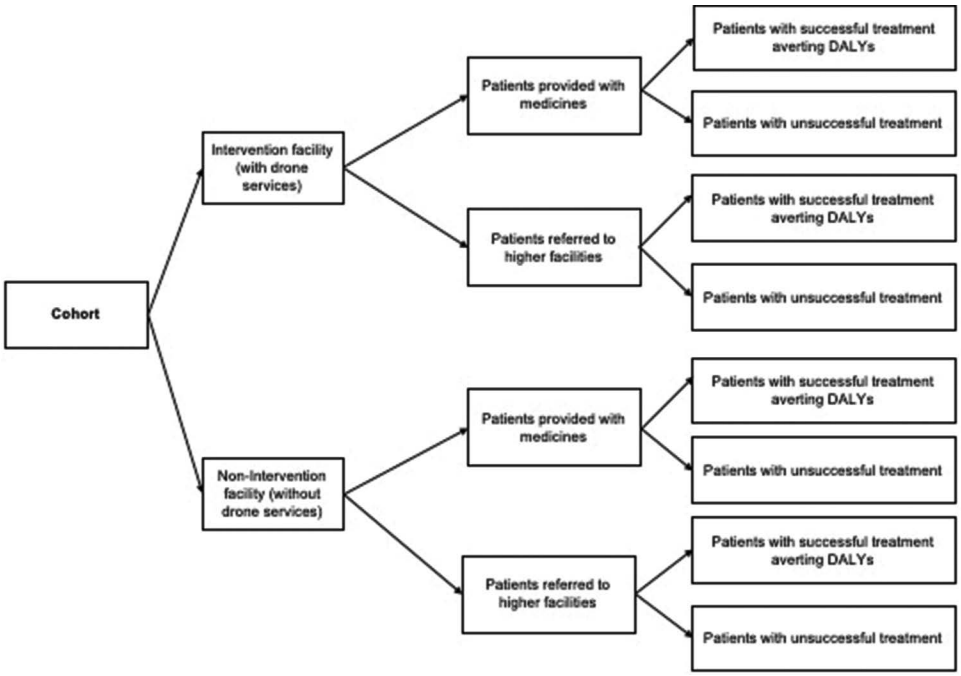

Figure 1: Decision tree model for the cost-effectiveness study

Assessment of carbon emission

The battery-operated drone technology has a niche over the other modes of transportation as the battery-operated drones are less producer of carbon footprint. Carbon emission is widely considered a crucial element of any rational climate strategy, serving as an efficient, adaptable, and reasonably priced means of curbing carbon emissions. We compared the total carbon emission between drone and vehicle for two sites. The carbon emission for the vehicle was calculated as per the CO_2 emission calculator.[8] CO_2 emission by drone was calculated using the following formula:

Total CO_2 emissions = Energy consumption × emission factor where, energy consumption (Wh) = Power consumption (W) × time (h)

Results

The Government of Meghalaya initiated the drone-based medical supplies under the MHSSP on December 5, 2022. The drone services connected 11 primary healthcare (PHC) facilities stepwise (initially 5 sites, later extended to 11 sites) in a hub and spoke model [Figure 2].

Using the distance chart method, the average land distance was mapped 49 km in the area of operation, Table 1. The list of all healthcare facilities (sites) connected by drones along with aerial and land distances and flight duration (time) is provided in Table 1. The remotest healthcare facility was Nongalbibra PHC, which was 123 km by road and took 02:25 h to reach by road, whereas the drone traversed an aerial distance of 44·36 km in 30 min to reach the Nongalbibra PHC. Further, no adverse event was reported during the study duration.

Table 1: Aerial and land distance between the hub and spokes

Hub name	Spokes	Average land distance (in km)	Average time by road (in h)	Aerial distance (in km)	Average time by drone (in h)
SDH - Jengjal	Pedaldoba PHC	64·8	2:00	37·6	0:25
	Nongalbibra PHC	123	2:25	44·36	0:30
	Shallang PHC	111	2:35	54·6	0:36
	Samanda PHC	37·9	0:55	18·2	0:12
	Mendipather PHC	59.2	1:40	44·4	0:29
	Asanang PHC	18	0:35	8·5	0:05
	Dadenggre CHC	42·2	1:20	15·5	0:10
	Gabil PHC	20·2	0:50	13·4	0:09
	Manikganj PHC	62·2	1:45	48	0:30
	Dobu PHC	70·3	2:20	40	0:26
	Tikrikilla PHC	52	1:10	36·25	0:24

SDH=Sub-Divisional Hospital, PHC=Primary Healthcare Centres

A total of 214 flights (one-way flights) were conducted transporting 199·05 kg of payload to 11 spoke sites from December 2022 to January 2024. The total number of flights included the trial/test flights undertaken to the healthcare facility [Figure 3a]. Also, the decrease in the number of flights from June 2023 to November 2023 was due to the high number of rainy days during the corresponding period [Figure 3b and c]. Road transportation during the rainy days was equally challenging.

Cost comparison between vehicle and drone-based transportation

A comparative costing analysis between vehicle-based and drone-based transportation was conducted to evaluate and

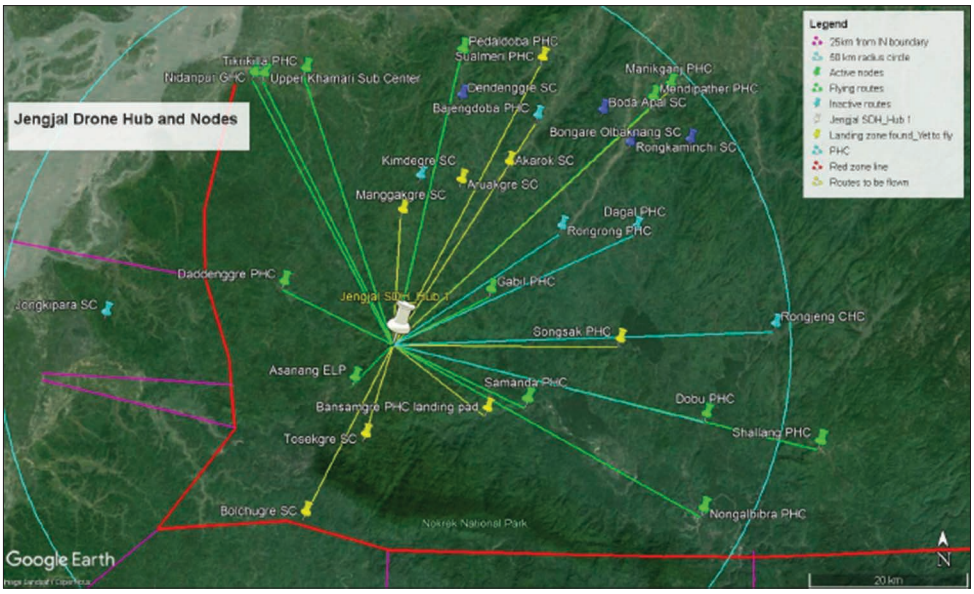

Figure 2: Drone delivery Hub and Spoke Model in Meghalaya

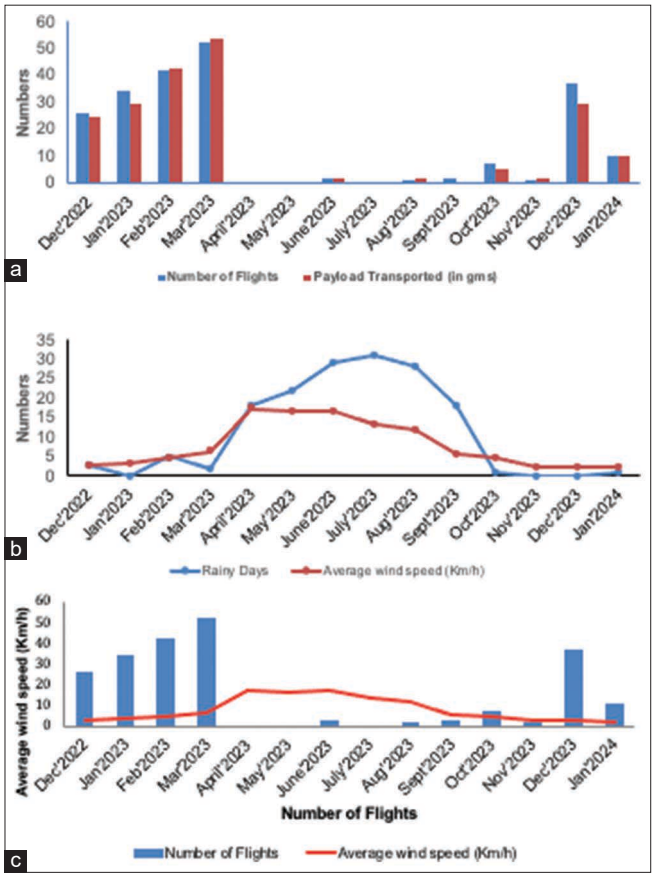

Figure 3: (a) Number of flights and total weight of payload delivered by drone from December 2022 to January 2024. The total number of flights included the trial/test flights undertaken to the healthcare facility. (b) Average number of rainy days per month and average wind speed in Meghalaya from December 2022 to January 2024. (c) Drone flyability during high wind velocity

quantify the differences in operational and economic efficiency from the healthcare provider's perspective [Figure 4 and Table 2].

Table 2: Total cost for vehicle and drone-based transportation of medical logistics per round trip between the Hub (Sub Divisional Hospital - Jengjal) and Spoke (Primary Healthcare Centres Pedaldoba)

Cost	Costing details	Average cost for vehicle-based transportation (INR)	Average cost for drone-based transportation (INR)
CapEx	Cost (annualized)	245	1282
OpEx	Utility cost	1113·6	1128
	Maintenance cost (including insurance cost)	329	
	Average HR cost	1214	1333
Total cost*		2902	3743

*Total cost for vehicle transportation=Capital cost + Recurrent cost, where capital cost is the vehicle cost and recurrent cost is the utility cost+maintenance cost + human resource cost

The analysis included a detailed estimation of various cost components such as the capital cost of procurement, recurrent, maintenance, and human resources costs. This comprehensive assessment enabled an in-depth understanding of the financial implications associated with each transportation mode, accounting for both upfront investments and ongoing operational expenditures. By incorporating these critical cost factors, the study aimed to provide a clear comparison of the cost-effectiveness and sustainability of vehicle versus drone logistics solutions within the medical supply chain.

Costing for vehicle-based transportation

Capital cost

Capital cost is defined as a fixed, one-time expense incurred on the purchase of necessities used in rendering services. The vehicle being used for transportation of logistics was TATA SUMO (Model 2016). Considering inflation rates for India since 2016, the present cost of a vehicle was found to be INR

765,000 (including insurance cost). The annualized cost (AF of 0·117) for vehicle cost was found to be INR 89,500 (considering a life expectancy of 10 years and a discount rate of 3%). The cost incurred by the health system for vehicles is INR 89,500 per year. Therefore, considering the utilization of a vehicle in a year for transportation of medical consumables, the capital cost of the vehicle was found to be approximately INR 245 per round trip. The comparison was conducted for the conventional service on the same day as the drone service.

Recurrent cost

Recurrent costs are the costs of maintaining and operating a given program or procedure. For the vehicle transportation of medical logistics between the SDH (Jengjal) and the PHC, several recurrent costs were identified and included in the calculation, including:

Utility cost

The utility cost of the vehicle included its diesel fuel consumption, which was costed at an average INR 87 per liter. The scheduled trip for transportation of medical logistics through vehicle between the SDH and the PHC was once in a month. The average estimated distance traveled in a round trip by the vehicle between the SDH and PHC was 128 km. The mileage of the vehicle was 10 km per liter of fuel and was calculated to be INR 1113.6 per round trip for transportation of medical logistics through road transport.

Maintenance cost

For the study purpose, the estimated maintenance cost of the vehicle was assumed to be on a higher bracket of INR 10,000 per month due to the hilly terrain. No paper record of the actual maintenance expenditure was available at the healthcare facility. The average maintenance cost for the vehicle was INR 329 per round trip.

Human resource cost

The cost of human resources was calculated considering the actual time spent on the activity per month. One vehicle/ambulance driver and one loader were considered as human resources involved. The salaries of the driver and loader per month were INR 25,000 and INR 11,430, respectively, and were considered for the costing exercise (as per the Meghalaya public health system and State labor wages norms). Therefore, the human resource cost was INR 1214 per round trip.

Costing for drone-based transportation

Capital cost

The only capital cost that was required for this transportation was the drone cost. At present, the drone cost is INR 4,000,000 (including insurance). The annualized cost (AF of 0·117 using a life expectancy of 10 years, a discount rate of 3%) for drone cost was found to be INR 468,000 per year. Therefore, considering the utilization of drone in a year for transportation of consumables, the capital cost of the drone is approximately INR 1282 per round trip.

Recurrent cost

Recurrent costs are the costs of maintaining and operating a given program or procedure. For the drone transportation of medical logistics between the SDH (Jengjal) and the PHC, several recurrent costs were identified and included in the calculation, including:

Utility cost and maintenance cost

The utility cost of the drone included its electricity usage for charging the batteries and minor repairs. We assumed 10% of the rate fixed by the Government for service charge (i.e. INR 150 per aerial km) and arrived at INR 15 per aerial km. The average aerial distance traveled in a round trip by the drone between the SDH and PHC was 75·2 km and covered it in 50 min. The utility cost and maintenance cost for 75·2 km was found to be INR 1128 for one round trip.

Human resource cost

The cost of human resources was calculated based on the monthly salary of all personnel involved. One drone pilot and one assistant were considered as human resources involved. The salaries of one drone pilot and one assistant per month were INR 25,000 and INR 15,000, respectively, and were considered for the costing exercise. Therefore, the human resource cost was INR 1333 per round trip.

Time-related outcomes

To assess the benefits of timely transportation of medical logistics, the time taken by drone transportation and vehicle transportation was recorded and compared. This is important in medical emergency cases.

1. Time taken for one round trip through vehicle transportation from Hub to Pedaldoga PHC = 240 min (4 h)
2. Time taken for one round trip through drone transportation from Hub to Pedaldoga PHC = 50 min.

Economic evaluation

Incremental cost-effectiveness ratio

The cost-effectiveness ratios (CERs) of both transportation modalities were calculated by dividing the Grand Total Procedure Cost by the outcome (time outcome/patient health outcome). The calculations of CER for both modes of transportation are shown below:

- CER1 of vehicle transportation = Total cost per trip/effectiveness of transportation through vehicle (effectiveness in terms of time and health outcomes)
- CER2 of drone transportation = Total cost per trip/effectiveness of transportation through drone (effectiveness in terms of time and health outcomes).

The economic findings were summarized by the analysis of the ICER. The ICER is defined as the difference in cost between two possible health interventions, divided by the difference in their effect or outcomes. Therefore, ICER = CER2-CER1

Formula for ICER calculation:

ICER = C1-C2/E1 − E2

where C1 = Cost of transportation through vehicle

C2 = Cost of transportation through drone

E1 = Effectiveness of transportation through vehicle

E2 = Effectiveness of transportation through drone.

CER1 for PHC Pedaldoba (cost-effectiveness ratio for vehicle transportation) = Cost incurred through vehicle/Time taken by vehicle

= INR 2902/240 min = INR 12·09/min

CER2 for PHC Pedaldoba (cost-effectiveness ratio for drone transportation) = cost incurred through drone/time taken by the drone

= INR 3743/55 min = INR 68·05/min

ICER = CER2-CER1 = INR 68·05–12·09 = INR 56/min

Therefore, for saving 1 minute of transport, INR 56 shall be spent through drone services.

Patient-related health outcomes

The health benefits gained by the patients receiving treatment from the intervention healthcare facility (PHC Pedaldoba) and comparator PHC Bajendoba were measured. Data were collected from facility records and interviews with healthcare staff at the facility (MO, pharmacist, nurse). The average number of patients seeking treatment in intervention healthcare facility (footfall) was around 1000 patients per month with antenatal patients being the majority cases (70%). Most of the emergency cases were for road traffic accidents, dog bites, pneumonia, hemorrhage cases, and high-risk pregnancies. On an average, 4–5 cases per month were referred to higher healthcare facilities for treatment. However, subsequent patient management was conducted at the PHC facility. The data were compared to healthcare facilities without drone services in terms of DALYs averted.

Case 1: Health facility having drone services

Disability Adjusted Life Years averted

Disability-adjusted life years are a measure of overall disease burden, expressed as the number of years lost due to ill health, disability, or early death.

Calculation of DALYs:
Number of DALYs = Years lived with disability (YLD) + years of life lost (YLL)
Number of DALYs = Disability weight x years lived with disability + (life expectancy – age at death)

Certain emergency medicines like rabies vaccine, anti-venom injection, injection Labetalol, Botroclot (Haemocoagulase) topical solution, injection Etophylline, injection Adrenaline, injection Dexamethasone, and injection Tranexamic Acid were provided in emergency during the past 1 year. This prevented referrals to district hospitals and saved lives, time, and funds. With timely treatment, many lives were saved and effectiveness was documented in the form of DALYs averted. As per health facility records, every month, on an average, two pneumonia cases and one dog bite case were treated by the medical supplies transported by drones. Hence, the two cases, Pneumonia and Dogbite, were selected for the DALYs. Taking the life expectancy for India to be 70 years and the disability weight for pneumonia being 0·3, the DALYs averted[9] with the timely treatment of pneumonia were 36 DALYs for each patient with an average age of 20 years. Likewise, the DALYs averted for one dog bite (assumption of disability weight being 0.2) case were 34 DALYs. In total, 106 DALYs were everted each month in the facility with drone services. Detailed DALY calculations are provided in Supplementary File 1.

Economic impact: Data from patients from both the intervention and comparator health facilities were collected to assimilate their economic burden (out-of-pocket expenditure) due to medical conditions. The expenses saved by averting referrals were assessed. The amount of money patients spent each time going to the higher facility was calculated with respect to the distance. The distance between PHC Pedaldoba and SDH Jengjal was 64·8 km, and the travel cost was INR 800 per patient. This amount was saved each time a patient averted referral and received treatment at the facility. Moreover, opportunity cost due to loss of wages (approximately INR 500/day) was also averted each time a referral was avoided. This amount was saved due to timely treatment in intervention facilities. On average, the three cases (two pneumonia and one dog bite case) treated at the facility led to savings of INR 2400 with respect to avoidance of referral and INR 1500 as avoidance of opportunity cost every month.

Time saved: The drone took 25 min to reach the PHC Pedaldoba and it took 12 minutes to reach the PHC Samanda, whereas the vehicle took 2 hours to reach PHC Pedaldoba and 50 minutes to reach PHC Samanda. This advantage of 38 to 95 minutes acts as a window of opportunity to save patients requiring emergency treatment. Delivery of logistics by drone helped in provision of timely treatment to emergency patients.

Case 2: Health facility without drone services

Disability Adjusted Life Years averted

The facility without drone services also provided emergency treatment. However, through record keeping, it was found that one pneumonia patient was treated every month, but the dog bite cases were referred to higher facilities. Therefore, in these facilities, only 36 DALYs were averted.

Figure 4: Comparison between drone-serviced healthcare facility and non-drone-serviced spoke sites

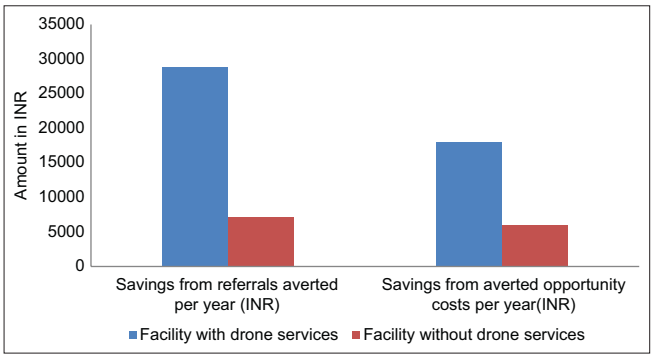

Figure 5: Comparison of monetary gains of intervention and comparator facilities

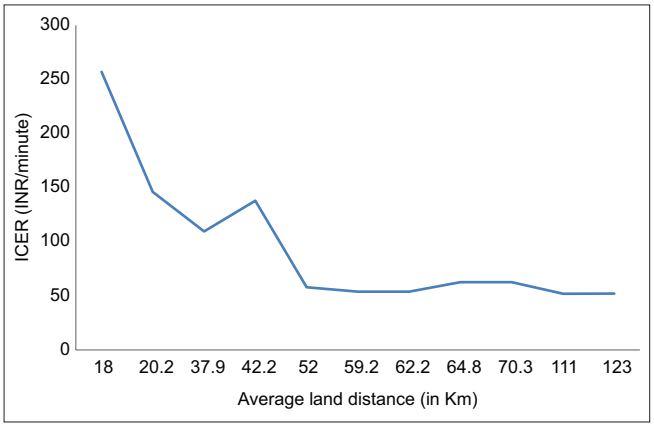

Figure 6: Effect of distance on Incremental Cost Effectiveness Ratio from the Hub (SDH Jengjal)

Economic impact: The distance between PHC Bajengdoba to SDH Jengjal was 61 km, and the travel cost was INR 600. This amount was saved each time a referral was averted for one patient each month. The opportunity cost due to loss of wages (approximately INR 500 per patient) was also averted for this patient. Notably, there were more referrals in the facility without drone services.

Economic evaluation

Cost-effectiveness ratio

The ICER is defined as the difference in cost between two possible health interventions, divided by the difference in their effect or outcomes.

Formula for ICER calculation:

ICER = C1-C2/E1-E2

where, C1 = Cost of transportation through drone

C2 = Cost of transportation through vehicle

E1 = DALYs averted through drone

E2 = DALYs averted through vehicle

ICER = C1-C2/E1-E2

ICER = 3743-2902/106-36 = 841/70

= INR 12/DALY averted.

Therefore, we need to spend an additional INR 12 to avert one DALY for the facility with drone services as compared to the facility without drone services. The ICER is well below the willingness to pay threshold that is GDP per capita for India (INR 200,030 for year 2022).

Net monetary gains in a year

The intervention saved lives and out-of-pocket expenditure leading to timely health outcomes. For graphical representation, the incidence of three patients' health outcome included in the study at the drone-serviced facility and one patient's health outcome in the non-drone health facility has been extrapolated for a 12-month period. The comparison of monetary gains with patients' perspective in a year with respect to the beneficiaries of the drone-serviced facility as compared to the non-drone-serviced health facility is shown in Figure 5.

Sensitivity analysis

Sensitivity analysis is a methodology to assess the impact, effect, or influence of key assumptions or variations (inputs or parameters in different case scenarios on the overall conclusions of a study (ICER in this study). In this study, the effect of distance of PHC on ICER (INR/min, extra funds need to be paid to gain 1 minute of time in case of emergency) will be worthwhile and have substantial value addition in the expansion of drone service delivery. The ICERs for variation in distance of PHC from the Hub were calculated from the information provided in Table 1. The variation of ICER was plotted in a graph along with the PHCs and their distance from the Hub [Figure 4]. The graph is given below. It concluded that the greater the distance of PHC from the SDH (drone service station), the lower is the ICER, confirming that drone services are more cost-effective in far and hard-to-reach areas [Figure 6]. The graph also depicts that drone is most cost-effective when the distance is more than 65 km and this finding is in congruence with a study done by Ochieng *et al.*, the finding of which also showed that vehicles were more cost-effective in the shorter distance (<65 km).[10]

Table 3: Comparison of carbon emission between drone and vehicle

	Carbon emission for drone	Carbon emission for vehicle
Hub (SDH Jengjal) to Spoke (Pedaldoba PHC)	Power consumption (W)=80 W Time (h)=0·4 h Energy consumption=32 Wh Emission Factor=0·5 kg CO_2/kWh	Energy consumption=7 L Emission factor=2·68kg CO_2 per L
Total emission	0·016 kg CO_2	18·76 kg CO_2

PHC=Primary Healthcare Centres

Drone versus vehicle carbon footprints

Although there are no direct emissions during drone delivery, the emissions computation included indirect carbon emissions in the power generation stage. The comparative analysis between the drone and a vehicle's carbon emission is provided in Table 3. The CO_2 emission per package delivered by drones was up to 99% less than the conventional mode of transportation.

Discussion

The use of drone technology in healthcare is a relatively new concept. Drones can give rich dividends in disasters and emergencies where timely delivery is a crucial component. Delays in medical supply delivery can be caused by a variety of issues, including inefficient transportation systems, heavy traffic, and unfavorable environmental conditions. By overcoming geographical barriers, reducing delivery times, and enhancing cost-effectiveness, drones can reshape the landscape of medicine delivery, particularly in remote and underserved areas.[11,12] The BVLOS category drones is a challenging domain; however, use of the drones can leapfrog the geographical hurdles and help in last-mile delivery of medical logistics in remotely located and in disaster situations when the land routes are cut off.[13]

The BVLOS category of drones has higher performance measures as compared to visual line-of-sight drones, used in the agriculture sector. Over the past few years, the role of drones in saving lives has already been established.[14] Haidari *et al.* reported improved vaccine availability by drones (96% vs. 94%) compared to traditional land transport. Moreover, drones produced cost savings of $0.08 per vaccine dose administered, thus leading to cost savings ranging from $0.05 to $0.21 per dose across different scenarios.[15] The drone technology complements the existing supply chain routes to overcome the delays/bottleneck and stockouts as a possible use case which is experienced in the Meghalaya project. The firming up of use case for regular delivery of drugs would require a curated inventory of emergency supplies and an online system of indenting by the healthcare facility as per their requirement.

It is important to record the demand generated from the spokes (PHC/CHC) as it will differ depending on the OPD/IPD caseloads. Improved availability of drugs/emergency medicine supplies using drones as additional transport means complementing the existing vehicle transportation at the healthcare facility will have a higher impact on healthcare outcomes and clientele satisfaction. Moreover, the delivery of blood components and temperature-sensitive drugs and diagnostics samples using drones can tremendously contribute to better health outcomes in future. Additionally, community engagement is crucial for efficient drone delivery.[16]

Integrated drone delivery for healthcare improvement (IDDI)

Drone has a lot of limitations, some of which need to be prioritized and addressed immediately more than others. Given the trade-off between weight, distance, and cost, UAV transport will likely supplement medical supply chains, rather than replace road transport. Drones are best suited for hard-to-reach areas/hilly and difficult terrains, where a conventional transportation system is often delayed or nonviable.

1. Prescheduling of drugs and diagnostics delivery: Prescheduling of drone-based delivery against the advance demand/indent received from the healthcare facility will help in improving the satisfaction factor and trust among the healthcare workers. The healthcare facility provides their monthly indents in advance for the drone service provider to schedule the flight as per the payload. Advance scheduling works interactively with multiple or no constraints and helps to generate, modify, update, or plan delivery schedules to meet the demand. Considering the high lead time for procurement to delivery, advance scheduling can reduce the number of stockout days.
2. Mitigating bad weather days: Drone-based deliveries are vulnerable to high wind velocity, temperature changes, and precipitation or other weather-related factors. The median flyability for the drones is low in a high wind velocity of >35–45 km/h. This hampers their reliability in time-sensitive operations.[17] A higher wind velocity in headwind can retard the drone's flying time due to increased consumption of battery power for the increased drag, and crosswinds leading to instability of the drone mid-flight. The absence of accurate, up-to-date meteorological data can create several operational and safety issues. Real-time weather apps or tools can help to stay updated on changes that could affect the drone's performance. Moreover, advance stockings should be kept by the healthcare facility.
3. Advanced Battery system: The BVLOS drones work on high-performance batteries (20,000–25,000 mAh) which require recharge and in due course are subjected to degradation which if not detected can jeopardize the drone's flight. The in-house capacity of manufacturing intelligent rechargeable batteries and their effective disposal is crucial for successful outcomes. This can be undertaken under the Make-in-India campaign with the Government encouraging production-linked initiatives to set up manufacturing units at par with international standards.

The limitations of the study were the nonavailability of individual flight data for a relatively longer period of time, as compared to the data used in the study. Nonavailability of

regular or cyclical fixed indents from the healthcare facility was a deterrent for planning scheduled flights. We overcame the limitation by considering the economic evaluation of drones with a comparable single round trip of the vehicle used. Second, only two health conditions were studied for arriving into the economic evaluation. From our own experience, it is difficult to obtain ethical approval not at the moment given the current challenge of the nonavailability of the BVLOS drone flight in large-scale regulatory approvals in the country.

Conclusion

Drones are increasingly being tested for healthcare purposes around the globe. It is a tool complementary to existing transport systems offering advantages over traditional approaches in specific circumstances. The ICER for drone transportation is INR 12 per DALY averted as compared to vehicle transportation of medical logistics. This is well below the willingness to pay threshold (GDP per capita of India). The healthcare facility having drone services was able to avert 106 DALYs compared to 36 DALYs per month by a healthcare facility with a conventional mode of transportation. The drone services would require an additional INR 56 to save 1 minute for the transport of medical logistics. Factors such as bad weather days (rainy days or high wind velocity), indenting schedule, drone endurance, payload capacity, and battery recharging hindered the adoption of drone technology in healthcare.

Drone technology for the supply of rescue medicine during emergencies, particularly in hilly and remote areas, can save lives and lead to improved health outcomes. Our findings highlight significant disruption to drone services during the bad weather days. How and where drones optimally fit into health systems is still being determined and will depend on local needs and resources. A comprehensive understanding and overcoming the barriers associated with drone technology can help to address supply chain shortcomings, reducing stockouts and wastage in the healthcare sector, and thus incorporate drone technology into the healthcare system.

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Conflicts of interest

There are no conflicts of interest.

References

1. Aggarwal S, Balaji S, Gupta P, Mahajan N, Nigam K, Singh KJ, *et al.* Enhancing healthcare access: Drone-based delivery of medicines and vaccines in hard-to-reach terrains of Northeastern India. Prev Med Res Rev 2024;1:172-8.
2. Eichleay M, Evens E, Stankevitz K. Parker C. Using the unmanned aerial vehicle delivery decision tool to consider transporting medical supplies via drone. Glob Health Sci Pract 2019;7:500-6.
3. Aggarwal S, Gupta P, Mahajan N, Balaji S, Singh KJ, Bhargava B, *et al.* Implementation of drone based delivery of medical supplies in North-East India: Experiences, challenges and adopted strategies. Front Public Health 2023;11:1128886.
4. Zailani MA, Azma RZ, Aniza I, Rahana AR, Ismail MS, Shahnaz IS, *et al.* Drone versus ambulance for blood products transportation: An economic evaluation study. BMC Health Serv Res 2021;21:1308.
5. Thakur V, Ganeshkumar P, Lakshmanan S, Rubeshkumar P. Do unmanned aerial vehicles reduce the duration and costs in transporting sputum samples? A feasibility study conducted in Himachal Pradesh, India. Trans R Soc Trop Med Hyg 2022;116:971-3.
6. Raykar NP, Raguveer V, Abdella YE, *et al.* Innovative blood transfusion strategies to address global blood deserts: A consensus statement from the blood delivery via emerging strategies for emergency remote transfusion (Blood DESERT) coalition. Lancet Glob Health 2024;12:e522-9.
7. World Weather Online. Available from: https://www.worldweatheronline.com/shillong-weather-averages/meghalaya/in.aspx. [Last accessed on Mar 2024 06].
8. NITI Aayog CO2 emission calculator. Available from: https://e-amrit.niti.gov.in/co2-calculator. [Last accessed on 2024 Mar 06].
9. Global Burden of Disease; 2021. Available from: https://healthdata.org/research-analysis/gbd. [Last accessed on 2024 March 06].
10. Ochieng WO, Ye T, Scheel C, Lor A, Saindon J, Yee SL, *et al.* Uncrewed aircraft systems versus motorcycles to deliver laboratory samples in west Africa: A comparative economic study. Lancet Glob Health 2020;8:e143-51.
11. Gupta P, Choudhury R, Kotwal A. Achieving health equity through healthcare technology: Perspective from India. J Family Med Prim Care 2023;12:1814-7.
12. Johannessen KA. A conceptual approach to time savings and cost competitiveness assessments for drone transport of biologic samples with unmanned aerial systems (Drones). Drones 2022;6:62.
13. Sharma S, Sharma H. Drone a technological leap in health care delivery in distant and remote inaccessible areas: A narrative review. Saudi J Anaesth 2024;18:95-9.
14. Schierbeck S, Nord A, Svensson L, Ringh M, Nordberg P, Hollenberg J, *et al.* Drone delivery of automated external defibrillators compared with ambulance arrival in real-life suspected out-of-hospital cardiac arrests: A prospective observational study in Sweden. Lancet Digit Health 2023;5:e862-71.
15. Haidari LA, Brown ST, Ferguson M, Bancroft E, Spiker M, Wilcox A, *et al.* The economic and operational value of using drones to transport vaccines. Vaccine

2016;34:4062-7.

16. Gupta P. At the Crossroads for the Use of Drones in Public Health System in India: Leveraging Community Mobilisation. Indian J of Community Med 2025 :10.4103/ijcm.ijcm_446_24. | DOI: 10.4103/ijcm.ijcm_446_24.
17. Gao M, Hugenholtz CH, Fox TA, Kucharczyk M, Barchyn TE, Nesbit PR. Weather constraints on global drone flyability. Sci Rep 2021;11:12092.

Supplementary File 1

(i) CALCULATIONS OF DALY FOR PNEUMONIA CASE

DALY for pneumonia case = Disability Weight x Number Of Years Lived With Disability+ Life Expectancy – Age At Death

(The Disability weights are typically estimated on a scale from 0 (no health loss) to 1 (equivalent to death). The disability weight for pneumonia being 0·3 the DALYs averted)[1]

= 0.3*20 Years Lived With Disability + 70 Years (Life Expectancy) – 40 Years (Age At Death)

=.6+30=36 Years

(ii) CALCULATIONS OF DALY FOR DOG BITE CASE

DALY For Dog Bite Case = 0.2*20 Years Lived With Disability + 70 Years – 40 Years

(The disability weight for dog bite being 0·2 the DALYs averted)[1]

= 0.2*20 Years Lived With Disability + 70 Years (Life Expectancy) – 40 Years (Age At Death)

= 4+30 = 34 Years

[1] Global Burden of Disease https://healthdata.org/research-analysis/gbd